

Sudiksha Chindula

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B.Tech in Computer Science, 2023-2027, PES University | CGPA: 8.56

Experience

Software Developer Intern Hyderabad, India
AVEVA *Jun 2026 – Present*

- Built a frontend widget integrating a live data streaming pipeline using **Angular**, **TypeScript**, **Highcharts**, and **GraphQL** to visualise complex factory data streams for process improvement.
- Deployed and maintained the feature branch on Microsoft Azure, managing the build for a live, testable environment.

Research Intern Bangalore, India
Centre for Information Security, Forensics and Cyber Resilience *Jun 2025 – Jul 2025*

- Built a **novel unsupervised, multi-modal pipeline** to detect anomalous actions in videos from the NWPU Campus Dataset and provide context summaries along with a three-tier adaptive summarization algorithm and a composite anomaly scoring mechanism.
- Integrated **YOLOv11** for object detection, **optical flow** for motion analysis, **Llava-Next** for vision-language captioning, and **Llama3** for summarisation achieving BERTScore similarity up to **0.90** and **75.48%** anomaly classification accuracy.

Teaching Assistant, Operating Systems Bangalore, India
PES University *Jan 2026 – May 2026*

- Designed and evaluated graded assignments, and authored new course content for the 2028 batch covering **process management**, **concurrency**, and **systems programming** which revamped the prior technical depth of the course’s assignments.

Projects

Custom Implementations from Scratch C

- **grep**: Supports regex pattern matching across files and directories along with standard grep flags.
- Docker’s build and runtime engine: Includes a CLI, an image-layer build system, and a container runtime.
- Redis server in C: Implements the RESP protocol, concurrent client handling, and core commands.

Distributed Data Marketplace Platform Apache Kafka, MySQL, Flask, REST API

- Built a 4-node distributed data marketplace enabling producers to publish data products and consumers to subscribe to them, with a central control plane managing approval workflows and metadata-driven routing.
- Designed a multi-threaded producer that dynamically provisions topics via the admin.

CNN Accelerator on FPGA PyTorch, Zynq, Python

- Trained and quantized a CNN for digit classification, converting floating-point weights to fixed-point integer arithmetic for deployment on a **Zynq** FPGA.

Scrum Call Assistant Python, Sentence-Transformers, Pinecone

- Built an Agentic RAG chatbot that reduces redundancy in daily scrum calls. Employees can log their updates onto a database that can be queried by managers through a chat interface to get updates asynchronously.

Research Publications

Pharmacokinetic State Space Models for Unbiased Prediction of Haemodynamic Collapse
AI in Healthcare Conference @ Imperial College, London (Accepted, Nominated for Best Paper) 2026

- Proposed a **Mamba-based** model predicting IOH events from drug infusion history, achieving **AUROC 0.7360** and a **2.73-fold** AUPRC lift over baseline with constant-memory scaling versus Transformer alternatives.

Additional Information

Programming Languages: Python, C, SQL, Java, HTML/CSS, Angular, JavaScript, TypeScript.
Technologies: Git, Docker, CUDA, Autopsy, Apache Kafka, Apache Hadoop, MySQL, Redis, Pinecone, MongoDB.
Certifications: Agentic Systems 101, PESU I/O, Slot 18: Advanced RAG systems, AI Agentic Architecture.
National-level Debater and President, PES DebSoc: Won Open Best Speaker and Best Judge across India.
Scholarships: Recipient of the CNR scholarship, awarded to the top 20% of the batch.